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Chemistry 12

Lab 3A: Precipitation Reactions

1 Purpose

To determine an empirical value of K_{sp} for PbI_2. This experiment was performed with two different procedures. Both will be covered.

2 Materials

- 0.010M $Pb(NO_3)_2$ solution
- 0.020M KI solution
- Erlenmeyer flask (>100mL)
- Graduated cylinders (10mL & 50mL)
- Burette

2.1 Alternative Materials

- 0.010M $Pb(NO_3)_2$ solution
- 0.020M KI solution
- Beaker (>100mL)
- Hot air oven
- Filter paper

3 Procedure

- Dilute 10.0mL of 0.010M $Pb(NO_3)_2$ solution to 50.0mL (0.0020M) with distilled water. Transfer to the Erlenmeyer flask.
- Pour 0.020M KI solution into the burette
- Titrate the KI solution into the $Pb(NO_3)_2$ solution until a precipitate forms.
- Record ΔV of KI. This will be used to calculate K_{sp}.

3.1 Alternative Procedure

- Mix 10.0mL of 0.010M $Pb(NO_3)_2$ solution with 10.0mL of 0.020M KI solution in a beaker.
- Let mixture sit to allow precipitate to form.
- Filter precipitate out of solution and dry in hot air oven
- Measure mass of dried precipitate. This will be used to calculate K_{sp}.

4 Data/Observations

Both solutions started as clear, colourless liquids. As KI was added to the $Pb(NO_3)_2$ solution, the mixture became cloudy, then started forming small crystals at the bottom. Finally, a thin yellow hue developed. ΔV of the KI was recorded at each step.

Table 1: ΔV and visual observations during the test.

ΔV (mL)	Visual
0.0	Clear solution
5.0	Cloudy solution
5.9	Small crystals forming on bottom
9.0	Extremely cloudy solution
12.0	Tiny hue developing
14.8	Thin yellow hue developing

Table 2: Parameters.

Parameter	Value
PbI_2 Molarity	0.0020M
KI Molarity	0.020M
PbI_2 Volume	50.0mL

(a) ΔV = 5.9 mL. (b) ΔV = 12.0 mL. (c) ΔV = 14.8 mL.

Figure 1: Solution at various levels of ΔV . Note that the sparkles are not visible in photo a, but were observed in person.

4.1 Alternative Data/Observations

Table 3: Alternative Data (raw).

Parameter	Mass (g)
m (filter ppt)	2.55g
m (filter avg)	2.47g
m (ppt)	0.08g

Table 4: Alternative Data (averages).

Parameter	Mass (g)
Filter 1	2.41g
Filter 2	2.55g
Filter 3	2.40g
Filter 4	2.58g
Filter 5	2.41g
Filter avg	2.47g

Both solutions started as clear, colourless liquids. When they were mixed, a bright yellow precipitate formed immediately. This was then filtered out and dehydrated in the hot air oven. The dehydrated precipitate appeared as a bright yellow powder. The mass of the precipitate + filter paper was recorded, as well as an an average value for the mass of the filter paper alone.

Because we forgot to weigh the filter paper before filtering, we had to use an average value over five filters. Using this value, we can calculate the mass of the precipitate alone:

$$m_{ppt} = m_{filter,ppt} - m_{filter,avg}$$
$$m_{ppt} = 2.55g - 2.47g$$
$$m_{ppt} = 0.08g$$

5 Analysis

Using the ΔV value we can calculate an estimated K_{sp} for PbI_2. We will use the ΔV when small crystals of PbI_2 started appearing on the bottom of the flask. This is because the Ksp value represents when the solution is just saturated, and barely any precipitate is formed.

The K_{sp} can be calculated via:

$$K_{sp} = [\mathrm{Pb}^{2+}][\mathrm{I}^-]^2 \tag{1}$$

When we add KI to the solution, the volume of the liquid increases, diluting both solutions. Therefore we must factor this into our calculations.

Using $c_1v_1 = c_2v_2$, we can find the new concentrations of Pb^{2+} and I^- after x mL of 0.020M KI is added:

$$[\mathrm{Pb}^{2+}] = 0.0020M \cdot \frac{50.0mL}{50.0mL + x\ mL} \tag{2}$$

$$[\mathrm{I}^-] = 0.020M \cdot \frac{x\ mL}{50.0mL + x\ mL} \tag{3}$$

Using this we can model K_{sp} as a function of x (with two significant figures):

$$K_{sp}(x) = \left(0.0020M \cdot \frac{50.0mL}{50.0mL + x\ mL}\right) \cdot \left(0.020M \cdot \frac{x\ mL}{50.0mL + x\ mL}\right)^2 \tag{4}$$

When $\Delta V = 5.9$ mL, $K_{sp}(5.9\ mL) = 8.0 \times 10^{-9}$

We can view this graphically as well:

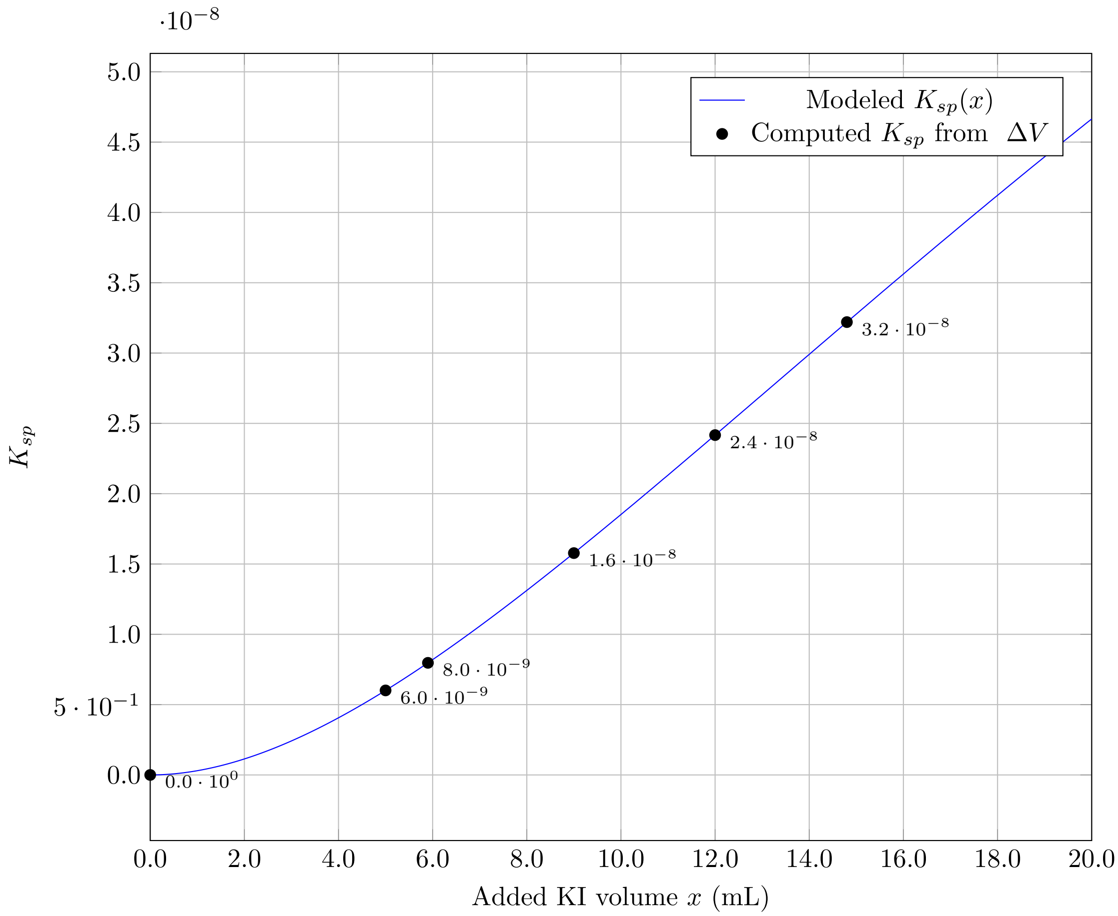

Figure 2: Modeled $K_{sp}(x)$ for PbI_2 with measured ΔV points from Table 1.

5.1 Alternative Analysis

Using the calculated mass of the precipitate after drying, we can calculate the moles of PbI_2 that were formed, and thus the moles of Pb^{2+} and I^- that were removed from the solution. The remaining moles dissolved in the solutioncan be used to calculate a K_{sp} value.

Moles in precipitate (moles removed from solution):

$$0.08g\ \mathrm{PbI_2} \cdot \frac{1\ mol\ \mathrm{PbI_2}}{461.0g\ \mathrm{PbI_2}} \cdot \frac{1\ mol\ \mathrm{Pb^{2+}}}{1\ mol\ \mathrm{PbI_2}} = 2 \times 10^{-4}\ mol\ \mathrm{Pb^{2+}}$$
$$0.08g\ \mathrm{PbI_2} \cdot \frac{1\ mol\ \mathrm{PbI_2}}{461.0g\ \mathrm{PbI_2}} \cdot \frac{2\ mol\ \mathrm{I^-}}{1\ mol\ \mathrm{PbI_2}} = 3 \times 10^{-4}\ mol\ \mathrm{I^-}$$

Initial moles in solution ($n = cv$):

$$0.010M\ \mathrm{Pb(NO_3)_2} \cdot \frac{1M\ \mathrm{Pb^{2+}}}{1M\ \mathrm{Pb(NO_3)_2}} \cdot 10.0mL \cdot \frac{1L}{1000.0mL} = 1.0 \times 10^{-4}\ mol\ \mathrm{Pb^{2+}}$$
$$0.020M\ \mathrm{KI} \cdot \frac{1M\ \mathrm{I^-}}{1M\ \mathrm{KI}} \cdot 10.0mL \cdot \frac{1L}{1000.0mL} = 2.0 \times 10^{-4}\ mol\ \mathrm{I^-}$$

Here we already see the problem: we somehow ended up with more moles of Pb^{2+} and I^- in the precipitate than we started with in the solution. Because you cannot make matter out of thin air, this is impossible. We will cover possible sources of error in the conclusion.

6 Conclusion

The K_{sp} for PbI_2 is approximately 8.0x10^{-9} as calculated above. This was calculated with the datapoint taken at $\Delta V = 5.9$mL, which was when faint crystals started forming in the solution. This was when the solution was just barely saturated, as opposed to later measurements where enough precipitate formed to give the solution a yellow hue.

K_{sp} represents the maximum amount of ions that can be dissolved in a solution, therefore this datapoint was chosen as nearly all the ions were still dissolved with barely any precipitate, meaning that the calculations would be as close as possible to a perfectly saturated solution.

Had a later measurement been chosen, the calculated K_{sp} would have been much higher than the accurate value, as the ions forming the precipitate would still be included in the calculation. This would have resulted in an erroneously high solubility, telling us that PbI_2 could dissolve more ions than it actually can.

Similarly, had an earlier measurement been chosen, the calculated K_{sp} would tell us that PbI_2 has much lower solubility than it actually does and would not be able to dissolve as many ions as it actually can. This is because the solution would not have been saturated yet, meaning that more ions could still be dissolved.

6.1 Sources of Error

The actual K_{sp} is 8.5x10^{-9}, with our calculated value of 8.0x10^{-9} , our percent error is:

$$\frac{|8.0 \times 10^{-9} - 8.5 \times 10^{-9}|}{8.5 \times 10^{-9}} \times 100\% \approx 6\%$$

There are a couple sources of error in this experiment.

The limiting factor was the fact that we were observing whether a precipitate formed with our eyes. This meant that we were limited by our ability to see the faint crystals forming. Given a more precise tool, a more precise threshold could be found for when the solution was saturated. This could possibly be measured through conductance of the solution or faint colour detection.

Combined with the more precise tool, a burette with a smaller volume increment would aid in finding a more precise threshold. A smaller volume increment would help with creating a more precise starting value for the $Pb(NO_3)_2$ solution.

6.1.1 Alternative Sources of Error

This experiment was a failure, as we extracted more ions than existed in the original solutions. There are a couple of possible reasons for this. The first is that PbI_2 ions were not the only species captured by the filter paper. It is possible that some ${NO_3}^-$ or K^+ ions were captured by the filter paper, adding to the perceived mass of the precipitate.

The second reason is that the filter paper has variation in mass. Looking at the measured masses of the filter paper:

Table 5: Alternative Data (averages).

Parameter	Mass (g)
Filter 1	2.41g
Filter 2	2.55g
Filter 3	2.40g
Filter 4	2.58g
Filter 5	2.41g
Filter avg	2.47g

There is high variation in the mass of the filter paper, with a range of 0.18g. While the precipitate was measured to be 2.55g - 2.47g = 0.08g, it very much could have been 0.01g or 0.15g, which would have resulted in either a physically possible mass or a even less sensible value. This is further exacerbated by the fact that the scales used only had two decimal places of precision, meaning even less accuracy in the mass of the precipitate.